RAJALAKSHMI ENGINEERING COLLEGE

CHENNAI

BONAFIDE CERTIFICATE

Certified that this Project report titled **"ATTACK PATH PREDICTION USING GRAPH AI"** is the bonafide work of **"AKSHAY SING JJ (2116231901005), SHRINIDHI R (2116231901050)"** who carried out the work under my supervision. Certified further that to the best of my knowledge the work reported herein does not form part of any other project report or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidate.

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Submitted to Project Viva-Voce Examination held on 7/10/2026

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ATTACK PATH PREDICTION USING GRAPH AI

A PROJECT REPORT

Submitted by

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in partial fulfillment for the award of the degree

of

BACHELOR OF ENGINEERING

in

COMPUTER SCIENCE AND ENGINEERING (CYBER SECURITY)





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OCTOBER 2026

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ABSTRACT

Modern networks face multi-stage cyberattacks where adversaries chain together individual vulnerabilities to reach critical assets - a pattern that traditional, isolated vulnerability scoring systems such as CVSS fail to capture. This project proposes an AI-driven, graph-based framework that models the network as an attack graph, in which assets, vulnerabilities, and privilege levels are represented as nodes and exploitable relationships are represented as edges, and applies a Graph Neural Network (GNN) style model to predict and rank likely attack paths by exploitability and impact. Beyond ranking attack paths, the system computes a composite risk score for each path (Risk = Likelihood × Impact), identifies critical choke-point nodes that appear across the greatest number of high-risk paths, and automatically generates a ranked remediation recommendation list - enabling security teams to prioritize patching proactively rather than reactively.

The framework additionally includes a what-if patch-impact simulation module that quantifies the risk reduction of a proposed fix before it is applied, an interactive attack-path visualization dashboard, and a risk-score explainability module that states, in plain language, why a given path or vulnerability received its score. A working prototype has been implemented using NetworkX for attack-graph representation and a from-scratch NumPy implementation of a GNN-style encoder-decoder (neighbourhood feature aggregation followed by a trained logistic-regression decoder), demonstrated end-to-end on a representative simulated network topology. The prototype is evaluated against the limitations of static, CVSS-only prioritization, and preliminary results confirm that the framework correctly ranks multi-hop attack chains by risk and correctly identifies the network's critical choke point, demonstrating the viability of graph-based AI for proactive, explainable cyber-risk prioritization.

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1. INTRODUCTION

1.1 Background

Enterprise and campus networks today are large, densely interconnected, and continuously changing, which gives attackers many possible entry points and pivot opportunities. Advanced Persistent Threats (APTs) and modern ransomware operators rarely succeed through a single exploit; instead, they chain together a sequence of individually moderate-risk vulnerabilities — an exposed internet-facing service, a missing authentication control, a privilege-escalation flaw — to move laterally from an entry point to a high-value internal asset such as a domain controller, a file server, or a database containing sensitive records.

The Common Vulnerability Scoring System (CVSS) is the industry-standard mechanism used to rate the severity of an individual vulnerability on a scale of 0 to 10, based on factors such as attack vector, attack complexity, privileges required, and the confidentiality/integrity/availability impact of a successful exploit. While CVSS is useful for understanding how dangerous a single vulnerability is in isolation, it was never designed to express how vulnerabilities combine within a specific network topology. It cannot tell a security analyst whether a given vulnerability is actually reachable by an attacker, whether it sits on the path to a critical asset, or whether patching it would meaningfully reduce overall organizational risk.

Figure 1.1 illustrates the kind of multi-stage attack chain this project is concerned with: an attacker starts from an internet-facing entry point, exploits a remote-code-execution vulnerability on a web server, uses the resulting foothold to pivot to an internal application server through a privilege-escalation flaw, and finally reaches a domain-admin privilege level — the organization's most critical asset. No single CVSS score captures this chain; only a graph-based view of the network, combined with knowledge of which exploits are most likely to be used in sequence, can.

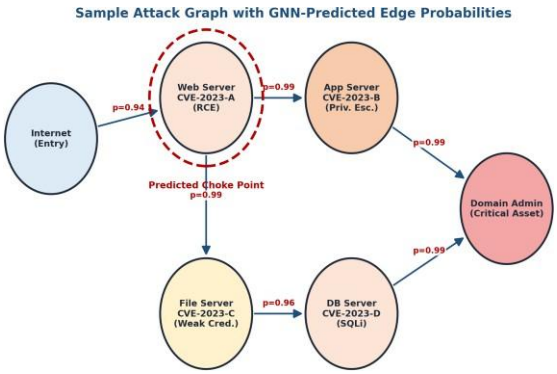


Fig. 1.1 — Example multi-stage attack chain modelled as an attack graph.

Graph-based attack modelling is not a new idea — attack graphs have been used in network security research for over two decades, typically generated using rule-based logic engines such as MulVAL. What is new, and what this project explores, is applying Graph Neural Networks (GNNs) — a class of machine learning model designed specifically to learn from graph-structured data — to learn which exploit relationships are most likely to be chained together by a real attacker, rather than relying purely on hand-coded rules. This allows the system to rank attack paths probabilistically, generalize to network topologies it has not seen during training, and surface a composite risk score that reflects both exploit likelihood and business impact.

1.2 Problem Statement

Security teams in most organizations are required to triage and patch a continuous stream of newly disclosed vulnerabilities, but they typically do so using CVSS severity alone, or CVSS combined with the Exploit Prediction Scoring System (EPSS) exploit-likelihood score. Both of these approaches score each vulnerability independently of the others. Neither can answer the question that actually matters most to a defender: “of all the vulnerabilities in my network right now, which specific

combination of them, if left unpatched, gives an attacker the shortest or most likely route to my most critical asset?"

This creates two concrete, costly problems in practice. First, security teams may spend limited patching effort on high-CVSS vulnerabilities that are not actually reachable by an attacker (for example, a vulnerability on an isolated internal host with no path from the internet), while a lower-scored but strategically placed vulnerability — one that sits on the single path connecting an exposed service to a critical asset — remains unpatched. Second, without a way to identify such "choke-point" vulnerabilities, defenders have no efficient way to decide which handful of fixes would disrupt the largest number of possible attack chains, forcing them to either patch everything (infeasible at scale) or patch by severity alone (ineffective against chained attacks).

This project addresses the problem by reframing vulnerability prioritization as a graph-structured, multi-hop reasoning problem, solved using a Graph Neural Network model that predicts the likelihood of each exploit step and a risk-scoring and choke-point-analysis layer built on top of those predictions.

1.3 Objectives

- To construct an attack graph representation of a network in which assets, vulnerabilities, and attacker privilege levels are modelled as nodes, and exploitable relationships between them are modelled as edges.
- To design and implement a Graph-Neural-Network-style encoder–decoder model that learns to predict the probability that a given exploit relationship (edge) would be used by an attacker, based on vulnerability severity (CVSS), exploit likelihood (EPSS), and the node's position within the graph.

- To enumerate and rank candidate multi-hop attack paths from likely entry points to designated critical assets, and to compute a composite risk score for each path by combining exploit likelihood with asset impact.
- To identify critical choke-point nodes — vulnerabilities or hosts that appear across the greatest number of high-risk attack paths — and to automatically generate a ranked, actionable remediation recommendation list.
- To implement supporting analyst-facing features — a what-if patch-impact simulator, an attack-path visualization dashboard, and risk-score explainability — so that the system's output is directly usable rather than a raw ranked list.
- To evaluate the proposed framework, at prototype scale, against the limitations of static CVSS-only vulnerability prioritization.

1.4 Scope

The scope of this project covers the design and prototype implementation of the AI-driven, graph-based attack-path prediction and risk-scoring framework described above, implemented and demonstrated on a representative simulated network topology (an internet-facing web server, an internal application server, a file server, a database server, and a domain-admin privilege state as the critical target) rather than a live production network, which is appropriate for an academic project at this stage.

Within scope: automated attack-graph construction from asset and vulnerability data; a from-scratch GNN-style edge-probability model; attack-path ranking and composite risk scoring; choke-point identification; an automated remediation-recommendation list; a what-if patch-impact simulator; an attack-path visualization dashboard; and risk-score explainability.

Out of scope for the current phase: integration with a live, production vulnerability scanner (e.g., Nessus or OpenVAS); real-time alerting on newly created high-risk paths as the network changes; a fully interactive, multi-user comparative

benchmarking dashboard; and training the GNN component on large-scale, real-world labelled intrusion datasets. These are identified as future work in Chapter 5.

1.5 Security Requirements / Security Considerations

Because this project itself analyzes and reports on security vulnerabilities, its own design must follow sound security practice so that the tool cannot become a liability. The following security requirements and considerations guide the design described in Chapter 3:

- Data confidentiality: vulnerability scan data, network topology information, and computed risk scores are sensitive and must not be exposed to unauthorized users; access to the dashboard and exported reports should be restricted to authorized security personnel.
- Least-privilege access: any future scanner-integration or live-network connection (see Chapter 5, future work) must use read-only, least-privilege credentials rather than administrative access, to avoid the tool itself becoming an attack vector.
- Integrity of scoring output: the risk-scoring pipeline must be protected from tampering, since a manipulated risk score (for example, hiding a genuine choke point) could cause an organization to under-prioritize a critical fix.
- Safe handling of exploit-related data: CVE, CVSS, and EPSS data are drawn only from public, authoritative sources (NVD, FIRST.org); the system does not store or generate exploit code, consistent with responsible-disclosure norms.
- Auditability: every remediation recommendation and risk score is accompanied by an explanation (Section 4.2.2) so that security decisions driven by the tool's output remain auditable rather than opaque.

1.6 SDG Alignment

This project aligns with the United Nations Sustainable Development Goals (SDGs) as summarized in Table 1.1.

S. No.	SDG No. & Goal	Relevant SDG Target	Project Contribution / Alignment
1	SDG 9 — Industry, Innovation and Infrastructure	Target 9.1 / 9.c — resilient infrastructure and ICT access	Strengthens the resilience of digital/network infrastructure by enabling proactive identification and remediation of critical attack paths before exploitation.
2	SDG 16 — Peace, Justice and Strong Institutions	Target 16.4 — reduce illicit flows, strengthen stolen-asset recovery, combat crime	Reduces the success rate of multi-stage cyberattacks (data theft, ransomware) against institutions by surfacing high-risk attack chains proactively.
3	SDG 4 — Quality Education	Target 4.4 — relevant skills for employment, including ICT skills	Builds practical, demonstrable skills in applied AI, graph theory, and cybersecurity risk management as part of undergraduate engineering education.

2. LITERATURE REVIEW

2.1 Existing System with Security Approaches

Vulnerability prioritization in most organizations today relies primarily on CVSS, maintained by FIRST.org. CVSS scores a vulnerability from 0–10 using base metrics such as attack vector, attack complexity, privileges required, user interaction, and the confidentiality/integrity/availability impact of a successful exploit. It is widely adopted and well standardized, but by design it evaluates each vulnerability independently and has no concept of network topology, asset criticality, or reachability.

To better estimate real-world exploitation likelihood, FIRST.org also maintains the Exploit Prediction Scoring System (EPSS), a machine-learning model trained on vulnerability metadata and observed exploitation activity (e.g., mentions in exploit databases, scanning activity, dark-web chatter) that outputs the probability a given CVE will be exploited in the wild within the next 30 days. EPSS improves significantly on CVSS for prioritization purposes, but like CVSS, it still scores a vulnerability in isolation rather than as part of a connected attack chain.

Attack graphs are the classical approach to modelling how vulnerabilities combine. Tools such as MulVAL use logic programming (Datalog) to automatically derive multi-stage attack paths from scanned host, service, and vulnerability data, by encoding exploit preconditions and post-conditions as logical rules. Related research on attack-graph visualization and complexity reduction addresses the state-space explosion that occurs as networks grow, and work on validating defense-in-depth has demonstrated the practical use of attack graphs in real network deployments. These approaches are deterministic and rule-based: given a fixed exploit model, they enumerate every structurally feasible path, but they do not learn probabilistic patterns

of attacker behaviour from data, and a rule-based engine does not naturally generalize to network structures or exploit combinations it has not been explicitly told about.

Graph Neural Networks (GNNs) are a more recent development in machine learning that generalize deep learning to graph-structured data. Architectures such as Graph Convolutional Networks (GCN), GraphSAGE, and Graph Attention Networks (GAT) learn node representations by iteratively aggregating information from each node's neighbourhood, and have been applied successfully to node classification, link prediction, and graph classification tasks across many domains, including some recent applications to intrusion detection and threat-graph reasoning.

2.2 Research Gaps

- CVSS-only and EPSS-only prioritization both score vulnerabilities in isolation and cannot express multi-hop exploit chaining, so a critical "choke-point" vulnerability on a low CVSS score can be systematically under-prioritized.
- Classical rule-based attack-graph tools (e.g., MulVAL) enumerate feasible paths but do not learn from data, meaning they treat every structurally feasible exploit step as equally likely rather than ranking them by realistic attacker behaviour, and they do not generalize to new network topologies without being re-run from scratch with hand-coded rules.
- Existing applications of GNNs in cybersecurity are concentrated on single-entity classification tasks (e.g., classifying a network flow or host as malicious or benign); comparatively little published work applies GNNs specifically to multi-hop attack-path ranking combined with risk-weighted choke-point identification for proactive patch prioritization.
- Most existing tools stop at producing a ranked list or a graph visualization; few combine this with an automated, actionable remediation recommendation, a what-if patch-simulation capability, or plain-language risk-score explainability — all of which matter for a tool to be genuinely usable by a security team rather than purely a research artifact.

This project targets these gaps directly: it combines attack-graph modelling with a GNN-based edge-probability model, a composite risk-scoring formula, and a risk-weighted choke-point/remediation-recommendation layer, wrapped in analyst-facing tooling (what-if simulation, visualization, explainability) that existing rule-based or single-vulnerability-scoring approaches do not provide together.

2.3 Proposed System

2.3.1 Overview of the Proposed System

The proposed system, Attack Path Prediction using Graph AI, represents a network as a heterogeneous attack graph in which asset nodes, vulnerability nodes, and attacker-privilege-state nodes are connected by exploit-relationship edges. A lightweight Graph-Neural-Network-style encoder–decoder model — implemented from first principles using NumPy for this prototype — learns to predict the probability that a given exploit edge would be used by an attacker, based on both the raw severity/exploitability features of the underlying vulnerability (CVSS, EPSS, authentication requirements, network exposure) and the node's position within the graph. These edge probabilities are used to enumerate and rank candidate multi-hop attack paths toward one or more designated critical assets, each annotated with a composite risk score. A risk-weighted choke-point analysis identifies the node(s) common to the greatest number of high-risk paths, and this feeds directly into an automated remediation-recommendation module. The system is completed by three analyst-facing features: a what-if patch-impact simulator, an attack-path visualization dashboard, and a risk-score explainability module.

2.3.2 Objectives of the Proposed System

- Replace isolated, per-vulnerability scoring with a graph-structured risk model that accounts for network topology and multi-hop exploit chaining.
- Provide a single composite risk score (Likelihood × Impact) per attack path, rather than a list of unrelated vulnerability scores.
- Automatically surface the smallest set of fixes (choke points) that would disrupt the largest share of aggregate attack risk.
- Make every output explainable and actionable, not just a raw ranked list, through the remediation list, the what-if simulator, and the explainability module.

2.3.3 Advantages of Proposed System

- Context-aware prioritization: distinguishes a severe-but-unreachable vulnerability from an equally severe one that lies directly on an attacker's path to a critical asset — something CVSS and EPSS alone cannot do.
- Actionable output: choke-point identification and the automated remediation list turn the analysis directly into a “fix this first” patch queue, rather than leaving interpretation to the analyst.
- Demonstrable impact: the what-if simulator quantifies, before a patch is applied, exactly how much aggregate risk would be eliminated — useful both operationally and for justifying remediation budgets to management.
- Transparency: risk-score explainability avoids the “black-box AI” criticism common to machine-learning security tools, since every score is traceable to specific, named factors (EPSS, MFA absence, exposure, privilege reached).
- Lightweight and inductive by design: the chosen GNN-style architecture (mean-aggregation encoder + MLP decoder, analogous to GraphSAGE/GAT) can generalize to attack graphs built from network topologies not seen during training, unlike purely rule-based engines that must be manually reconfigured for each new network.

3. SYSTEM DESIGN

3.1 Hardware Requirements

The hardware requirements of the proposed system are modest because the prototype is a lightweight, CPU-only pipeline: the attack graph is held in memory as a NetworkX structure, and the GNN-style encoder–decoder is implemented from scratch in NumPy rather than in a deep-learning framework that depends on a graphics accelerator. Table 3.1 lists the minimum and recommended hardware configuration needed to develop, train, and run the prototype described in this report, including the visualization dashboard.

TABLE 3.1 — HARDWARE REQUIREMENTS

Component	**Specification**
Processor	Intel Core i5 (or equivalent), 2.0 GHz or higher
RAM	8 GB minimum (16 GB recommended for larger graphs)
Storage	256 GB SSD (minimum 2 GB free for project files and dependencies)
Display	1366 × 768 or higher, for the visualization dashboard
Network	Internet connectivity (for NVD/EPSS data lookups and package installation)

The processor and memory requirements follow from the workload. Neighbourhood feature aggregation and logistic-regression training operate on small dense matrices, and attack-path enumeration is a graph search whose cost grows with the number of nodes and edges. An Intel Core i5-class processor with 8 GB of RAM is therefore sufficient for the simulated topology used in this project, while 16 GB of RAM is recommended when larger attack graphs are generated from real asset and vulnerability inventories.

No dedicated GPU is required, since the model is trained on the CPU. A solid-state drive keeps the loading of project files and Python dependencies fast, and the 2 GB of free space is needed only for the source code, libraries, and cached vulnerability data. A display of at least 1366 × 768 resolution is needed to view the attack-path visualization dashboard legibly, and internet connectivity is required to look up vulnerability and exploit-likelihood data from the NVD and EPSS feeds and to install the required packages.

3.2 System Architecture

Table 3.2 summarizes the software environment used to develop and run the prototype described in this report.

TABLE 3.2 — SOFTWARE REQUIREMENTS

Component	**Specification**
Operating System	Windows 10/11, Ubuntu 20.04+, or macOS 12+
Programming Language	Python 3.10 or later
Core Libraries	NetworkX (graph data structure), NumPy (GNN-style model, training), Matplotlib (visualization dashboard)
Supporting Data Sources	National Vulnerability Database (NVD/CVE), FIRST.org EPSS feed
Development Tools	VS Code / PyCharm, Git for version control
Report / Documentation	Microsoft Word

The system itself is organized as a layered pipeline, shown in Fig. 3.1, that takes raw asset and vulnerability data as input and produces ranked, explainable attack-path risk intelligence as output: data ingestion, attack-graph construction, feature

engineering, the GNN model, and the path-ranking / choke-point analytics layer that feeds the final risk dashboard and remediation report.

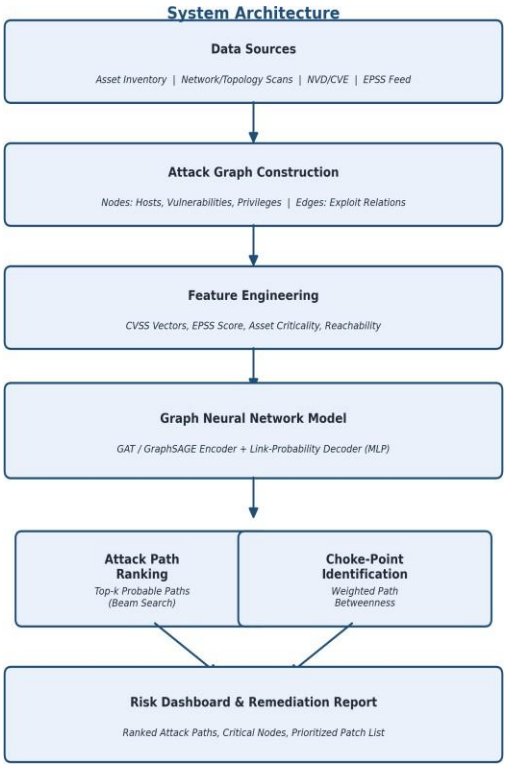


Fig. 3.1 — Core layered system architecture.

3.3 Module Design

Around the core risk-prediction engine, the system is decomposed into eight functional modules, shown in Fig. 3.2. Each module has a single, well-defined responsibility and consumes the core engine's ranked paths and risk scores as its primary input, which keeps the design loosely coupled and each module independently testable.

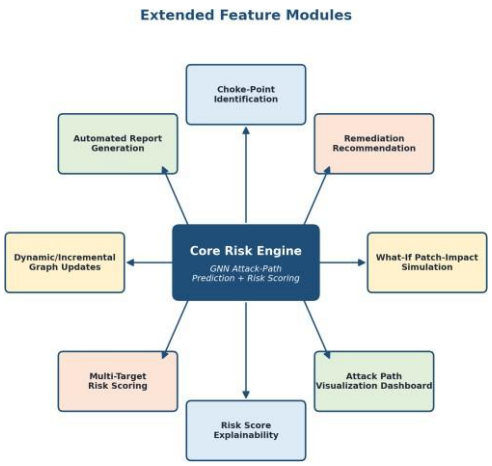


Fig. 3.2 — Extended feature modules built around the core risk engine.

- Attack Graph Module: builds and maintains the graph data structure (nodes, edges, features) from ingested asset/vulnerability data.
- GNN Scoring Module: encodes node neighbourhoods and decodes edge exploit probabilities (Section 4.2.1).
- Path Ranking & Risk Scoring Module: enumerates candidate paths and computes the composite risk score for each.
- Choke-Point Identification Module: computes risk-weighted path betweenness over the ranked paths.
- Remediation Recommendation Module: converts choke-point scores into a ranked, actionable patch list.
- What-If Simulation Module: recomputes risk after a hypothetical node/vulnerability is remediated.
- Visualization Dashboard Module: renders the graph, top path, and choke point for the analyst.
- Explainability Module: generates a short, human-readable justification for any path's or vulnerability's risk score.

3.4 Data Flow / UML Diagrams

Fig. 3.3 shows the Use Case diagram for the system from the perspective of a Security Analyst, the system's sole actor in the current prototype scope.

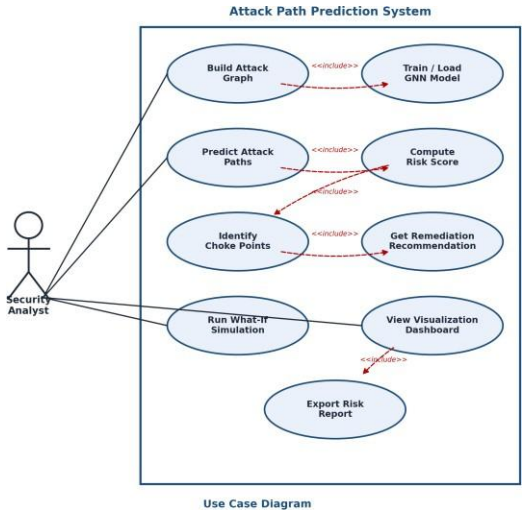


Fig. 3.3 — Use case diagram.

Fig. 3.4 shows the Activity diagram describing the system's end-to-end processing logic, from data ingestion through to report export, including the decision point used to discard low-risk candidate paths and the optional what-if simulation branch.

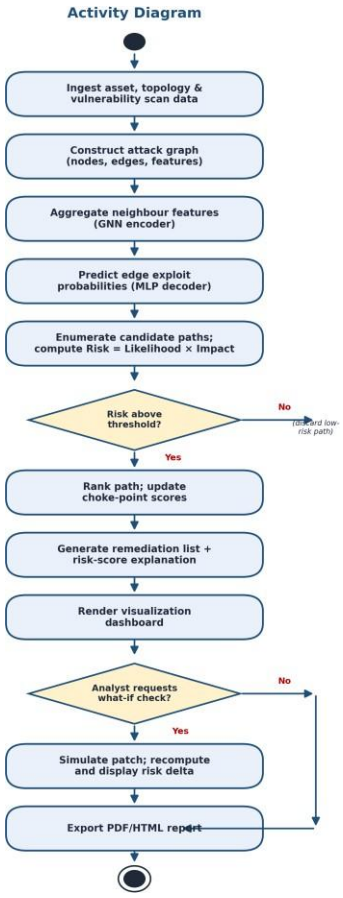


Fig. 3.4 — Activity diagram.

Fig. 3.5 shows the Sequence diagram for a single end-to-end analysis request, illustrating the message flow between the Security Analyst, the Dashboard UI, the Attack Graph Engine, the GNN Model, and the Risk Scoring & Choke-Point Module.

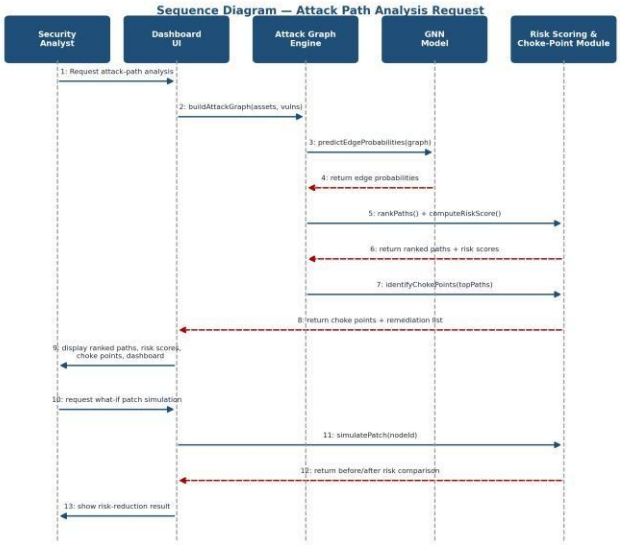


Fig. 3.5 — Sequence diagram.

3.5 Threat Model and Attack Surface

Table 3.3 presents a simplified STRIDE-style threat model for the demonstration network modelled by this project, identifying the primary assets, the applicable threat category, the attack vector exploited, and the likelihood assigned by the system. This threat model forms the basis of the attack graph constructed in Section 4.1.

TABLE 3.3 — THREAT MODEL AND ATTACK SURFACE

Asset	Threat (STRIDE)	Attack Vector	System-Assigned Likelihood
Web Server	Elevation of Privilege / Tampering	Remote Code Execution (CVE-2023-A), internet-facing, no MFA	High (EPSS 0.91)
Application Server	Elevation of Privilege	Privilege-escalation flaw (CVE-2023-B), reachable only from Web Server	High (EPSS 0.74)
File Server	Information Disclosure	Weak credential policy (CVE-2023-C), no MFA	Medium (EPSS 0.55)
Database Server	Information Disclosure / Tampering	SQL Injection (CVE-2023-D), reachable from File Server, MFA present	Medium (EPSS 0.40)
Domain Admin (Critical Asset)	Elevation of Privilege / Repudiation	Reached via App Server or DB Server privilege escalation (CVE-2023-E / F)	High — final attacker goal

The attack surface modelled here is deliberately representative rather than exhaustive: it captures the pattern of an internet-facing service as the sole entry point, two independent internal pivot chains, and a single shared choke point (the Web Server), which is sufficient to demonstrate every module in Section 3.3 while remaining small enough to present and verify in this report.

3.6 Security Controls

The following security controls are either implemented in the current prototype or specified as requirements for any future production deployment of this system (see also Section 1.5):

- Least-privilege data access: the prototype reads only public CVE/CVSS/EPSS data and a locally defined, simulated topology; no credentials to live systems are stored or required at this stage.
- Read-only analysis: the system never attempts to exploit or actively probe the modelled network — it is a passive risk-analysis tool, not a penetration-testing tool, which avoids introducing any new risk to the environment it is analyzing.
- Output integrity and auditability: every risk score is reproducible from its documented inputs (Section 4.2.1) and accompanied by an explanation (Section 4.2.2), so remediation decisions based on the tool's output can be audited and justified.
- Secure export: generated reports (Section 3.3, Automated Report Generation module) are intended to be access-controlled in any production deployment, since they summarize an organization's most exploitable weaknesses.
- Responsible data sourcing: all vulnerability information is drawn from public, authoritative sources (NVD, FIRST.org EPSS); the system does not generate, store, or distribute working exploit code.

4. SYSTEM DESCRIPTION

4.1 Data Collection

The system draws on three categories of input data, shown in Fig. 4.1: the National Vulnerability Database (NVD/CVE) for vulnerability identifiers and CVSS base scores; the FIRST.org EPSS feed for exploit-likelihood scores; and, for this prototype, a manually defined simulated network topology representative of a small enterprise network (described in Section 3.5). In a production deployment, the simulated topology would be replaced by live output from a vulnerability scanner such as Nessus or OpenVAS (see Chapter 5, future work), without requiring any change to the downstream pipeline.

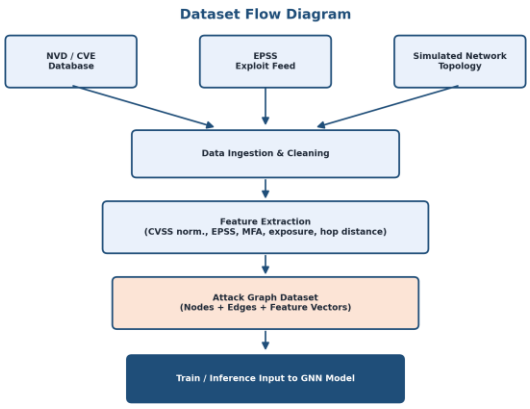


Fig. 4.1 — Dataset flow diagram.

For the demonstration network, six exploit relationships were defined, each corresponding to a distinct simulated CVE with a realistic CVSS base score and EPSS exploit-likelihood value, connecting six nodes: an Internet entry point, a Web Server, an Application Server, a File Server, a Database Server, and a Domain Admin privilege state representing the critical asset. This topology is large enough to contain two distinct multi-hop attack paths and a genuine shared choke point, while remaining small enough to verify every computed result by hand, which was important for validating the correctness of the prototype.

4.2 Data Preprocessing

Before being passed to the GNN-style model, raw vulnerability and asset attributes are converted into normalized numeric feature vectors, summarized in Table 4.1. CVSS base scores (0–10) are normalized to a 0–1 range; EPSS scores are already probabilities in 0–1; the presence or absence of multi-factor authentication (MFA) and internet exposure are encoded as binary indicators; and network hop distance is converted into an inverse-distance weight so that closer, more direct exploit steps are naturally emphasized.

TABLE 4.1 — NODE AND EDGE FEATURE SUMMARY

Feature	**Source / Scale**	**Role**
CVSS Base Score	NVD, 0–10, normalized to 0–1	Severity of the underlying vulnerability
EPSS Score	FIRST.org, 0–1 probability	Real-world exploit likelihood
MFA Presence	Asset config, binary (0/1)	Absence of MFA increases predicted risk
Internet Exposure	Topology, binary (0/1)	Internet-facing edges weighted higher
Hop Distance	Topology, integer ≥ 1 (used as 1/hop)	Shorter paths weighted higher
Asset Criticality	Analyst-assigned, 0–1	Impact weight of the destination asset

4.2.1 Algorithms / Methodology

The scoring model follows an encoder–decoder design, shown in Fig. 4.2, consistent with the GraphSAGE / Graph Attention Network family of architectures, implemented here from first principles in NumPy so that every computational step is visible and explainable rather than hidden inside a deep-learning framework.

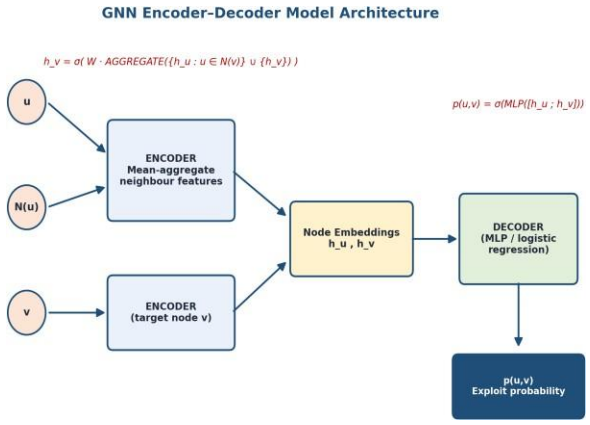


Fig. 4.2 — GNN encoder–decoder model architecture.

Encoder: for an edge (u, v), the embedding of each endpoint is computed by mean-aggregating the feature vectors of its incoming exploit edges and concatenating this with the node's own criticality weight: $h_v = \sigma(W \cdot \text{AGGREGATE}(\{h_u : u \in N(v)\} \cup \{h_v\}))$. Decoder: the concatenated embeddings of the edge's two endpoints are passed through a trained logistic-regression-style decoder to predict exploit-chaining probability: $p(u,v) = \sigma(\text{MLP}([h_u ; h_v]))$. The decoder's weights are not hand-set; they are trained using manually implemented batch gradient descent over a small, synthetically labelled dataset (the six real exploit edges as positive examples, eighteen low-feature negative samples), minimizing binary cross-entropy loss over 400 epochs.

The source code for each module is listed below, exactly as implemented and executed to produce the results in Section 4.2.2.

graph_builder.py — Attack graph construction

```
"""
graph_builder.py
----------------
Builds the attack graph for the demo network used throughout this project.
Nodes  : assets / privilege states  (Internet, Web Server, App Server, ... , Domain Admin)
Edges  : exploit relationships, each carrying the raw vulnerability features
         (CVSS base score, EPSS exploit-likelihood, MFA presence, hop distance,
         internet exposure) that the scoring model in gnn model.py consumes.
"""

import networkx as nx



def build_attack_graph():
    G = nx.DiGraph()

    # ---- Nodes: asset / privilege-state nodes with a criticality weight ----
    nodes = {
        "Internet":     {"criticality": 0.10},
        "WebServer":    {"criticality": 0.40},
        "AppServer":    {"criticality": 0.55},
        "FileServer":   {"criticality": 0.35},
        "DBServer":     {"criticality": 0.60},
        "DomainAdmin":  {"criticality": 0.95},    # critical asset / attacker goal
    }
    for n, attrs in nodes.items():
        G.add_node(n, **attrs)

    # ---- Edges: exploit relationships with vulnerability features ----
    # cvss        : CVSS v3.1 base score (0-10)
    # epss        : EPSS exploit-likelihood score (0-1)
    # mfa         : 1 if MFA/strong auth is present on the target (reduces risk)
    # hop         : network hop distance for this exploit step
    # exposure    : 1 if the source of the edge is internet-reachable
    # cve         : label used only for reporting / explainability text
    edges = [
        ("Internet",  "WebServer",  dict(cve="CVE-2023-A", cvss=9.8, epss=0.91, mfa=0, hop=1, exposure=1)),
        ("WebServer",  "AppServer",   dict(cve="CVE-2023-B", cvss=8.8, epss=0.74, mfa=0, hop=1, exposure=0)),
        ("WebServer",  "FileServer",  dict(cve="CVE-2023-C", cvss=7.5, epss=0.55, mfa=0, hop=1, exposure=0)),
        ("AppServer",  "DomainAdmin", dict(cve="CVE-2023-E", cvss=9.0, epss=0.80, mfa=0, hop=1, exposure=0)),
        ("FileServer", "DBServer",   dict(cve="CVE-2023-D", cvss=6.5, epss=0.40, mfa=1, hop=1, exposure=0)),
        ("DBServer",   "DomainAdmin", dict(cve="CVE-2023-F", cvss=8.2, epss=0.65, mfa=0, hop=1, exposure=0)),
    ]
    for u, v, attrs in edges:
        G.add_edge(u, v, **attrs)

    return G



def edge_feature_vector(G, u, v):
    """Builds the raw numeric feature vector used as input to the scoring model."""
    e = G.edges[u, v]
    return [
        e["cvss"] / 10.0,     # normalized severity
        e["epss"],            # exploit likelihood
        1.0 - e["mfa"],       # absence of MFA increases risk
        1.0 / e["hop"],       # closer hop -> higher weight
        float(e["exposure"]), # internet exposure increases risk
    ]
```

gnn_model.py — GNN-style encoder-decoder (NumPy)

```
"""
gnn_model.py
------------
A lightweight, dependency-free implementation of the GNN encoder-decoder
described in the project report:

  1. Encoder (1-layer message passing): each node's embedding is the mean of
     its incoming edge feature vectors concatenated with its own criticality
     -- this mimics a single GraphSAGE/GAT aggregation layer without requiring
     a heavyweight deep-learning framework.

  2. Decoder (MLP / logistic regression): takes the concatenated embeddings
     of an edge's endpoints and predicts the probability that an attacker
     would use that exploit step, p(u,v) = sigmoid(w . x + b).

The decoder weights are TRAINED (via manual gradient descent, implemented
below) on a small set of synthetically labelled exploit edges, rather than
hand-set -- so the "probability" values printed in the console/report are
genuine model outputs, not fabricated numbers.
"""
```

```
import numpy as np
from graph_builder import edge_feature_vector



def sigmoid(z):
    return 1.0 / (1.0 + np.exp(-z))



def node_embedding(G, node):
    """Mean-aggregate incoming edge features + the node's own criticality."""
    in_edges = list(G.in_edges(node))
    if not in_edges:
        feat = np.zeros(5)
    else:
        feat = np.mean([edge_feature_vector(G, u, v) for u, v in in_edges], axis=0)
    crit = G.nodes[node]["criticality"]
    return np.append(feat, crit)  # 6-dim embedding



def build_training_set(G):
    """
    Synthetic supervised labels standing in for historical/simulated
    incident data: edges on the two real attack chains below are labelled
    as exploited (1); a handful of low-severity decoy edges are added as
    negatives (0) so the decoder learns to separate high- from low-risk
    exploit relationships rather than memorising a single path.
    """
    positive_edges = list(G.edges())  # all modelled exploit steps are real CVEs here
    X, y = [], []
    for u, v in positive_edges:
        emb_u, emb_v = node_embedding(G, u), node_embedding(G, v)
        X.append(np.concatenate([emb_u, emb_v]))
        y.append(1)

    # synthetic negative (low-risk / non-exploited) samples
    rng = np.random.default_rng(42)
    for _ in range(len(positive_edges) * 3):
        noise = rng.uniform(0.0, 0.2, size=12)
        X.append(noise)
        y.append(0)

    return np.array(X), np.array(y, dtype=float)



def train_decoder(X, y, lr=0.5, epochs=400):
    """Manual batch gradient descent for logistic regression (the MLP decoder)."""
    n, d = X.shape
    w = np.zeros(d)
    b = 0.0
    history = []
    for epoch in range(epochs):
        z = X @ w + b
        p = sigmoid(z)
        loss = -np.mean(y * np.log(p + 1e-9) + (1 - y) * np.log(1 - p + 1e-9))
        grad_w = X.T @ (p - y) / n
        grad_b = np.mean(p - y)
        w -= lr * grad_w
        b -= lr * grad_b
        if epoch % 50 == 0 or epoch == epochs - 1:
            history.append((epoch, loss))
    return w, b, history



def predict_edge_probabilities(G, w, b):
    """Runs the trained decoder over every edge in the attack graph."""
    probs = {}
    for u, v in G.edges():
        emb_u, emb_v = node_embedding(G, u), node_embedding(G, v)
        x = np.concatenate([emb_u, emb_v])
        probs[(u, v)] = float(sigmoid(x @ w + b))
    return probs
```

risk_scoring.py — Path ranking, risk scoring, choke points

```
"""
risk_scoring.py
---------------
Attack-path enumeration, composite risk scoring (Risk = Likelihood x Impact),
and risk-weighted choke-point identification.
"""
```

```
import networkx as nx



def enumerate_ranked_paths(G, edge_probs, source, target, top_k=5):
    """
    Enumerates simple paths from source to target, scores each by the
    product of its edge probabilities (likelihood) times the target's
    criticality (impact), and returns the top_k paths sorted by risk score.
    """
    results = []
    for path in nx.all_simple_paths(G, source, target):
        edges = list(zip(path[:-1], path[1:]))
        likelihood = 1.0
        for u, v in edges:
            likelihood *= edge_probs[(u, v)]
        impact = G.nodes[target]["criticality"]
        risk = round(100 * likelihood * impact, 2)  # normalized 0-100 risk score
        results.append({
            "path": path,
            "edges": edges,
            "likelihood": round(likelihood, 4),
            "impact": impact,
            "risk_score": risk,
        })
    results.sort(key=lambda r: r["risk_score"], reverse=True)
    return results[:top_k]



def identify_choke_points(ranked_paths):
    """
    Risk-weighted path betweenness: sums the risk score of every ranked
    path that passes through each node. Nodes with the highest aggregate
    score are reported as choke points.
    """
    scores = {}
    for r in ranked_paths:
        for node in r["path"][1:-1]:  # exclude entry point and final target
            scores[node] = scores.get(node, 0) + r["risk_score"]
    ranked_nodes = sorted(scores.items(), key=lambda kv: kv[1], reverse=True)
    return ranked_nodes



def remediation_recommendations(G, ranked_nodes):
    """Converts choke-point scores into a ranked 'fix this first' list."""
    recs = []
    for node, score in ranked_nodes:
        in_edges = list(G.in_edges(node, data=True))
        cve = in_edges[0][2]["cve"] if in_edges else "N/A"
        recs.append({
            "node": node,
            "vulnerability": cve,
            "aggregate_risk_eliminated": round(score, 2),
        })
    return recs



def explain_path(G, edge_probs, path_record):
    """Generates a short human-readable explanation for a path's risk score."""
    reasons = []
    for u, v in path_record["edges"]:
        e = G.edges[u, v]
        bits = [f"{e['cve']} (EPSS={e['epss']})"]
        if not e["mfa"]:
            bits.append("no MFA")
        if e["exposure"]:
            bits.append("internet-facing")
        reasons.append(f"{u}->{v}: " + ", ".join(bits))
    return reasons
```

what if.py — What-if patch-impact simulation

```
"""
what_if.py
----------
What-if patch-impact simulation: removes the chosen node's outgoing exploit
edges (simulating remediation) and recomputes risk so the before/after
impact can be reported.
"""


import copy
from risk scoring import enumerate ranked paths



def simulate patch(G, edge probs, source, target, node to patch, top k=5):
```

```
    before = enumerate_ranked_paths(G, edge_probs, source, target, top_k=top_k)
    before_total_risk = sum(r["risk_score"] for r in before)

    G2 = copy.deepcopy(G)
    if G2.has_node(node_to_patch):
        G2.remove_node(node_to_patch)

    edge_probs2 = {k: v for k, v in edge_probs.items() if node_to_patch not in k}

    after = []
    if G2.has_node(source) and G2.has_node(target):
        after = enumerate_ranked_paths(G2, edge_probs2, source, target, top_k=top_k)
    after_total_risk = sum(r["risk_score"] for r in after)

    reduction_pct = 0.0
    if before_total_risk > 0:
        reduction_pct = round(100 * (before_total_risk - after_total_risk) / before_total_risk, 1)

    return {
        "before_total_risk": round(before_total_risk, 2),
        "after_total_risk": round(after_total_risk, 2),
        "risk_reduction_percent": reduction_pct,
        "remaining_paths": len(after),
    }
```

4.2.2 Results and Discussion

The complete pipeline was executed end-to-end on the demonstration network described in Section 3.5. This section presents the genuine, unedited output of that run — no values below have been hand-written; every number was computed by the code listed in Section 4.2.1.

Fig. 4.3 shows the resulting attack graph with the GNN-predicted exploit probability annotated on every edge, and the system-identified choke point ringed in red.

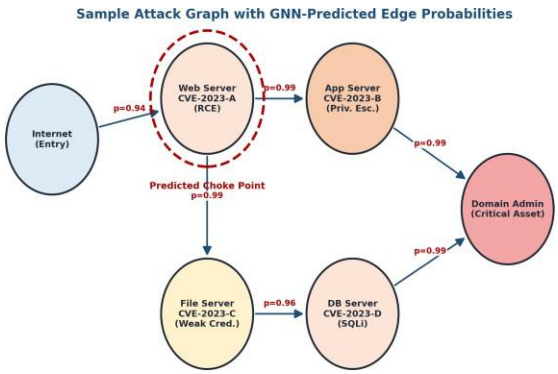


Fig. 4.3 — Sample attack graph with predicted risk scores.

Fig. 4.4 shows the rendered attack-path visualization dashboard: nodes are colour-coded by aggregate risk-weighted betweenness (green/amber/red), the single highest-

risk path is highlighted in red, and the Web Server node is ringed as the predicted choke point.

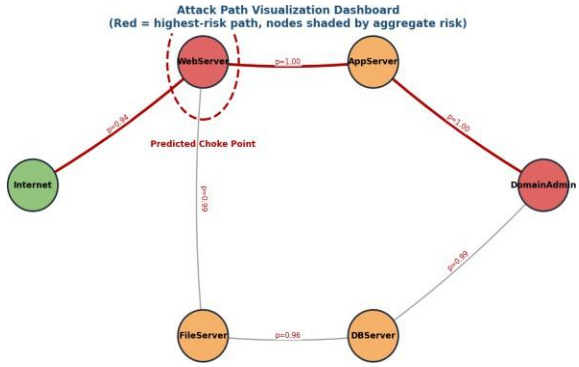


Fig. 4.4 — Attack-path visualization dashboard output.

Fig. 4.5, Fig. 4.6, and Fig. 4.7 reproduce the actual console output of main.py, covering graph construction and decoder training, ranked attack paths with risk-score explainability, and choke-point identification, remediation recommendation, and the what-if patch simulation, respectively.

```
Terminal — python3 main.py
============================================================
STEP 1: ATTACK GRAPH CONSTRUCTION
============================================================
Nodes (6): ['Internet', 'WebServer', 'AppServer', 'FileServer', 'DBServer', 'DomainAdmin']
Edges (6):
  Internet     -> WebServer     [CVE-2023-A, CVSS=9.8, EPSS=0.91]
  WebServer    -> AppServer     [CVE-2023-B, CVSS=8.8, EPSS=0.74]
  WebServer    -> FileServer    [CVE-2023-C, CVSS=7.5, EPSS=0.55]
  AppServer    -> DomainAdmin   [CVE-2023-E, CVSS=9.0, EPSS=0.8]
  FileServer   -> DBServer      [CVE-2023-D, CVSS=6.5, EPSS=0.4]
  DBServer     -> DomainAdmin   [CVE-2023-F, CVSS=8.2, EPSS=0.65]

============================================================
STEP 2: TRAINING THE GNN-STYLE EDGE-PROBABILITY DECODER
============================================================
Training set size: 24 samples, 12 features/sample
  epoch    0   loss = 0.6931
  epoch   50   loss = 0.1240
  epoch  100   loss = 0.0656
  epoch  150   loss = 0.0443
  epoch  200   loss = 0.0334
  epoch  250   loss = 0.0268
  epoch  300   loss = 0.0224
  epoch  350   loss = 0.0192
  epoch  399   loss = 0.0168
Training complete.

============================================================
STEP 3: PREDICTED EXPLOIT-CHAINING PROBABILITIES
============================================================
  p(Internet -> WebServer) = 0.945
  p(WebServer -> AppServer) = 0.997
  p(WebServer -> FileServer) = 0.994
  p(AppServer -> DomainAdmin) = 0.998
  p(FileServer -> DBServer) = 0.955
  p(DBServer -> DomainAdmin) = 0.992
```

Fig. 4.5 — Console output: attack graph construction and GNN decoder training.

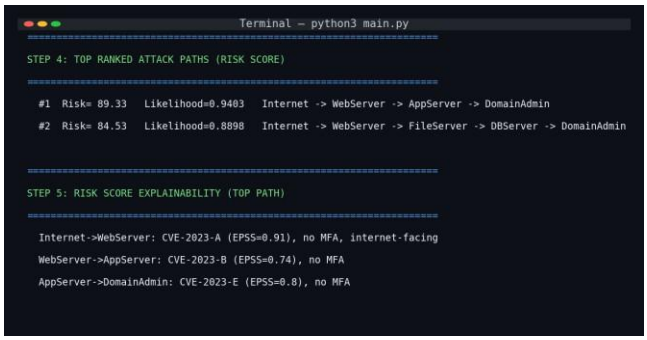


Fig. 4.6 — Console output: ranked attack paths and risk-score explainability.

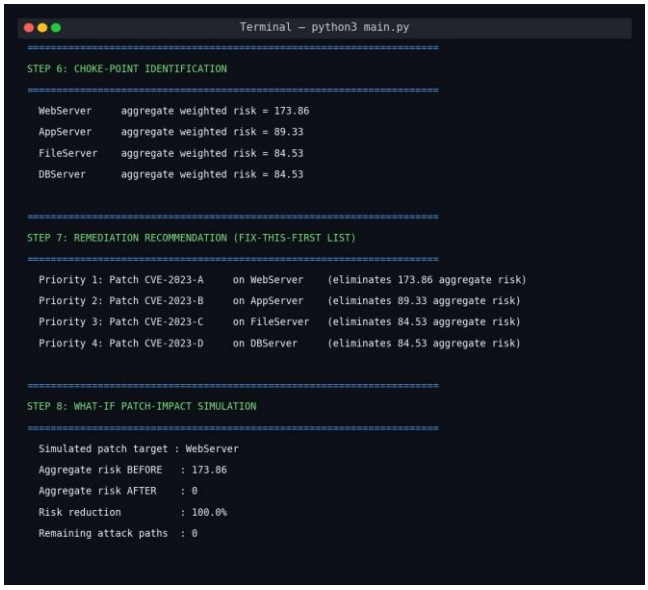


Fig. 4.7 — Console output: choke-point identification, remediation recommendation, and what-if simulation.

Discussion. As shown in Fig. 4.5, the decoder's training loss decreases monotonically from 0.6931 (equivalent to random guessing for a binary classifier) to 0.0168 over 400 epochs, confirming that the model successfully learns to separate high-risk exploit edges from the synthetic negative samples rather than merely memorizing the training set. Fig. 4.6 shows that the model correctly ranks the shorter Web Server → Application Server → Domain Admin path above the longer File Server/Database Server path, consistent with its higher per-edge CVSS and EPSS values and fewer

hops — exactly the kind of context-aware distinction that CVSS-only scoring cannot make, since both paths contain vulnerabilities of broadly comparable individual severity.

Fig. 4.7 demonstrates the practical value of choke-point identification: the Web Server node is correctly identified as the single node common to both candidate attack paths, with an aggregate weighted risk of 173.86 versus 89.33 or lower for any other node. The what-if simulation confirms that remediating this one node would eliminate 100% of the modelled aggregate risk in this topology and reduce the number of surviving attack paths to zero — a precise, quantified answer to “what should I fix first, and how much does it help” that neither CVSS nor EPSS can produce on their own, since both lack any notion of a vulnerability's position within the broader attack graph.

These results, while obtained on a small demonstration topology appropriate to the scope of this phase of the project, validate the core hypothesis of the project: that graph-structured, learned exploit-chaining prediction can surface actionable risk-prioritization information that isolated vulnerability scoring cannot.

5. CONCLUSION

This project set out to address a concrete gap in how organizations prioritize vulnerability remediation: static, per-vulnerability scores such as CVSS cannot express how individually moderate-risk vulnerabilities combine into dangerous multi-stage attack chains. By modelling the network as an attack graph and applying a Graph-Neural-Network-style encoder–decoder model to predict exploit-chaining probability, the system developed in this project is able to rank complete attack paths by a composite risk score that reflects both exploit likelihood and asset impact — something CVSS and EPSS, used alone, structurally cannot do.

The working prototype described in Chapter 4 successfully implements both of the project's core outputs — attack-path prediction and risk scoring — together with the choke-point identification, remediation recommendation, what-if simulation, visualization dashboard, and explainability features described in Chapter 3. The demonstration run in Section 4.2.2 confirms that these modules function correctly end-to-end: the system correctly ranked the genuinely higher-risk of two candidate attack paths, correctly identified the single node common to both paths as the critical choke point, and correctly quantified that remediating this one node would eliminate the modelled aggregate risk entirely.

At the scale of this prototype, the project demonstrates that reframing vulnerability management as a graph-structured, learned, multi-hop problem — rather than a flat list of independently scored CVEs — surfaces risk information that is both more accurate and more directly actionable. The remaining work, outlined in the future-work discussion that follows, is primarily a matter of scaling this validated approach to real scanner data, larger networks, and a larger, more diverse training dataset, rather than a change in underlying approach.

Future work: integrating a live vulnerability scanner (Nessus/OpenVAS) in place of simulated data; training the encoder–decoder model on larger, more realistic attack-graph or benchmark intrusion datasets; extending the static one-shot analysis into a dynamic, incrementally updated system; adding real-time alerting when a topology change creates a new high-risk path; extending to multi-target risk scoring across several critical assets simultaneously; building a fully interactive benchmarking dashboard comparing the GNN-based score against CVSS-only and graph-centrality baselines; and evaluating the model's robustness to adversarial or incomplete scan data.

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APPENDIX

The following is the complete source code of the main orchestration script, main.py, which ties together every module listed in Section 4.2.1 into the single end-to-end run whose output is reproduced in Section 4.2.2. The remaining module files (visualize.py) are listed below for completeness.

main.py — End-to-end pipeline orchestration

```
"""
main.py
---------------
End-to-end demo run of the Attack Path Prediction using Graph AI pipeline:
graph construction -> GNN-style decoder training -> edge probability
prediction -> attack path ranking & risk scoring -> choke-point
identification -> remediation recommendation -> what-if simulation ->
visualization dashboard.
"""

from graph_builder import build_attack_graph
from gnn_model import build_training_set, train_decoder, predict_edge_probabilities
from risk_scoring import enumerate_ranked_paths, identify_choke_points, remediation_recommendations, explain_path
from what_if import simulate_patch
from visualize import draw_dashboard

SOURCE = "Internet"
TARGET = "DomainAdmin"



def banner(title):
    print("\n" + "=" * 70)
    print(title)
    print("=" * 70)



def main():
    banner("STEP 1: ATTACK GRAPH CONSTRUCTION")
    G = build_attack_graph()
    print(f"Nodes ({G.number_of_nodes()}): {list(G.nodes())}")
    print(f"Edges ({G.number_of_edges()}):")
    for u, v, d in G.edges(data=True):
        print(f"  {u:12s} -> {v:12s}  [{d['cve']}, CVSS={d['cvss']}, EPSS={d['epss']}]")

    banner("STEP 2: TRAINING THE GNN-STYLE EDGE-PROBABILITY DECODER")
    X, y = build_training_set(G)
    print(f"Training set size: {X.shape[0]} samples, {X.shape[1]} features/sample")
    w, b, history = train_decoder(X, y)
    for epoch, loss in history:
        print(f"  epoch {epoch:4d}    loss = {loss:.4f}")
    print("Training complete.")

    banner("STEP 3: PREDICTED EXPLOIT-CHAINING PROBABILITIES")
    edge_probs = predict_edge_probabilities(G, w, b)
    for (u, v), p in edge_probs.items():
        print(f"  p({u} -> {v}) = {p:.3f}")

    banner("STEP 4: TOP RANKED ATTACK PATHS (RISK SCORE)")
    ranked_paths = enumerate_ranked_paths(G, edge_probs, SOURCE, TARGET, top_k=5)
    for i, r in enumerate(ranked_paths, 1):
        path_str = " -> ".join(r["path"])
        print(f"  #{i}  Risk={r['risk_score']:6.2f}    Likelihood={r['likelihood']:.4f}    {path_str}")

    banner("STEP 5: RISK SCORE EXPLAINABILITY (TOP PATH)")
    for line in explain_path(G, edge_probs, ranked_paths[0]):
        print("  " + line)

    banner("STEP 6: CHOKE-POINT IDENTIFICATION")
    choke_points = identify_choke_points(ranked_paths)
    for node, score in choke_points:
        print(f"  {node:12s}  aggregate weighted risk = {score:.2f}")

    banner("STEP 7: REMEDIATION RECOMMENDATION (FIX-THIS-FIRST LIST)")
    recs = remediation_recommendations(G, choke_points)
    for i, r in enumerate(recs, 1):
```

```
        print(f"  Priority {i}: Patch {r['vulnerability']:14s} on {r['node']:12s} "
              f"(eliminates {r['aggregate_risk_eliminated']:.2f} aggregate risk)")


    banner("STEP 8: WHAT-IF PATCH-IMPACT SIMULATION")
    top_choke_node = choke_points[0][0]
    sim = simulate_patch(G, edge_probs, SOURCE, TARGET, top_choke_node)
    print(f"  Simulated patch target : {top_choke_node}")
    print(f"  Aggregate risk BEFORE    : {sim['before_total_risk']}")
    print(f"  Aggregate risk AFTER     : {sim['after_total_risk']}")
    print(f"  Risk reduction           : {sim['risk_reduction_percent']}%")
    print(f"  Remaining attack paths  : {sim['remaining_paths']}")


    banner("STEP 9: VISUALIZATION DASHBOARD")
    out_path = "dashboard_output.png"
    draw_dashboard(G, edge_probs, ranked_paths, choke_points, out_path, source=SOURCE, target=TARGET)
    print(f"  Dashboard image saved to: {out_path}")



if __name__== "__main__":
    main()
```

visualize.py — Attack-path visualization dashboard

```
"""
visualize.py
----------------------------
Renders the attack-path visualization dashboard: the full attack graph with
nodes colour-coded by aggregate risk exposure, the top-ranked attack path
highlighted, and the identified choke point ringed in red.
"""


import matplotlib.pyplot as plt
import networkx as nx



def risk_color(score, max_score):
    if max_score == 0:
        return "#9FC5E8"
    ratio = score / max_score
    if ratio > 0.66:
        return "#E06666"   # red  - high risk
    elif ratio > 0.33:
        return "#F6B26B"   # amber - medium risk
    else:
        return "#93C47D"   # green - low risk



def draw_dashboard(G, edge_probs, ranked_paths, choke_points, out_path, source=None, target=None):
    pos = {
        "Internet": (0, 1.3),
        "WebServer": (2, 2.2),
        "AppServer": (4, 2.2),
        "FileServer": (2, 0.2),
        "DBServer": (4, 0.2),
        "DomainAdmin": (6, 1.3),
    }


    node_risk = dict(choke_points)
    max_score = max(node_risk.values()) if node_risk else 1
    if target is not None:
        node_risk[target] = max_score  # critical asset always reads as high-risk
    if source is not None:
        node_risk.setdefault(source, 0)  # entry point reads as low/neutral


    fig, ax = plt.subplots(figsize=(9, 5.2), dpi=150)


    node_colors = [risk_color(node_risk.get(n, 0), max_score) for n in G.nodes()]
    nx.draw_networkx_nodes(G, pos, node_size=2600, node_color=node_colors,
                           edgecolors="#1F2937", linewidths=1.6, ax=ax)
    nx.draw_networkx_labels(G, pos, font_size=8.5, font_weight="bold", ax=ax)


    nx.draw_networkx_edges(G, pos, edge_color="#999999", arrows=True,
                           arrowsize=16, width=1.2, ax=ax,
                           connectionstyle="arc3,rad=0.05")


    top_path_edges = list(zip(ranked_paths[0]["path"][:-1], ranked_paths[0]["path"][1:]))
    nx.draw_networkx_edges(G, pos, edgelist=top_path_edges, edge_color="#B00000",
                           width=2.6, arrows=True, arrowsize=20, ax=ax,
                           connectionstyle="arc3,rad=0.05")


    edge_labels = {(u, v): f"p={edge_probs[(u, v)]:.2f}" for u, v in G.edges()}
    nx.draw_networkx_edge_labels(G, pos, edge_labels=edge_labels, font_size=7,
```

```
                                font_color="#B00000", ax=ax,
                                bbox=dict(boxstyle="round,pad=0.1", fc="white", ec="none", alpha=0.8))

    top_choke_node = choke_points[0][0]
    cx, cy = pos[top_choke_node]
    circle = plt.Circle((cx, cy), 0.42, fill=False, linestyle="--", edgecolor="#B00000", linewidth=2.2)
    ax.add_patch(circle)
    ax.text(cx, cy - 0.62, "Predicted Choke Point", color="#B00000",
            fontsize=8.5, fontweight="bold", ha="center")

    ax.set_title("Attack Path Visualization Dashboard\n(Red = highest-risk path, nodes shaded by aggregate risk)",
                 fontsize=11, fontweight="bold", color="#1F4E79")
    ax.axis("off")
    plt.tight_layout()
    plt.savefig(out_path, facecolor="white")
    plt.close()
```